VICTORIA JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION 2025

H2 GEOGRAPHY **9173/01**
Paper 1

INSERT **3 hours**

READ THESE INSTRUCTIONS

The Insert contains all the Resources referred to in the questions.

Resource 1 for Question 1

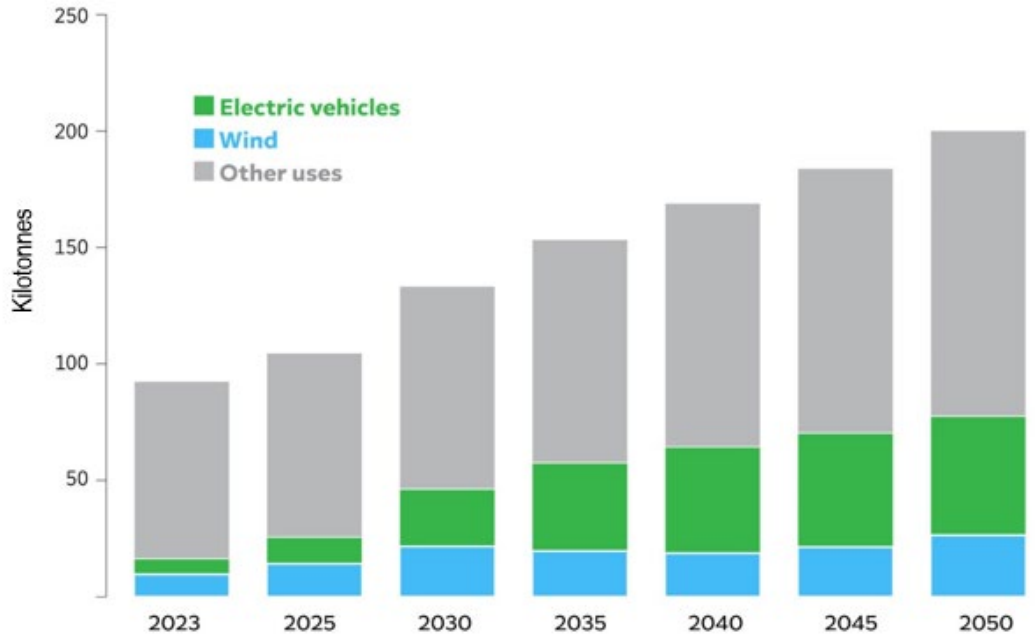
Global distribution of rare earth reserves and rare earth metals (REM) extracted in 2020

Country	REM extracted (tonnes)	Reserves (tonnes)	% of total reserves
China	140,000	44,000,000	38.0%
Vietnam	1,000	22,000,000	19.0%
Brazil	1,000	21,000,000	18.1%
Russia	2,700	12,000,000	10.4%
India	3,000	6,900,000	6.0%
Australia	17,000	4,100,000	3.5%
United States	38,000	1,500,000	1.3%
Greenland	-	1,500,000	1.3%
Tanzania	-	890,000	0.8%
Canada	-	830,000	0.7%
South Africa	-	790,000	0.7%
Other countries	100	310,000	0.3%
Burma	30,000	N/A	N/A
Madagascar	8,000	N/A	N/A
Thailand	2,000	N/A	N/A
Burundi	500	N/A	N/A
World Total	243,300	115,820,000	100%

REM: Rare earth metals
N/A: Data that is unavailable or undocumented

Resource 2 for Question 1

Trends in global demand for rare earth metals



Rare earth metals are used in the production of electric vehicles (EVs) and wind turbines.

Resource 3 for Question 1

News article highlighting the global rare earth metals shortage

<u>Controlled shortage of rare earth elements send chaos into global car industry</u>

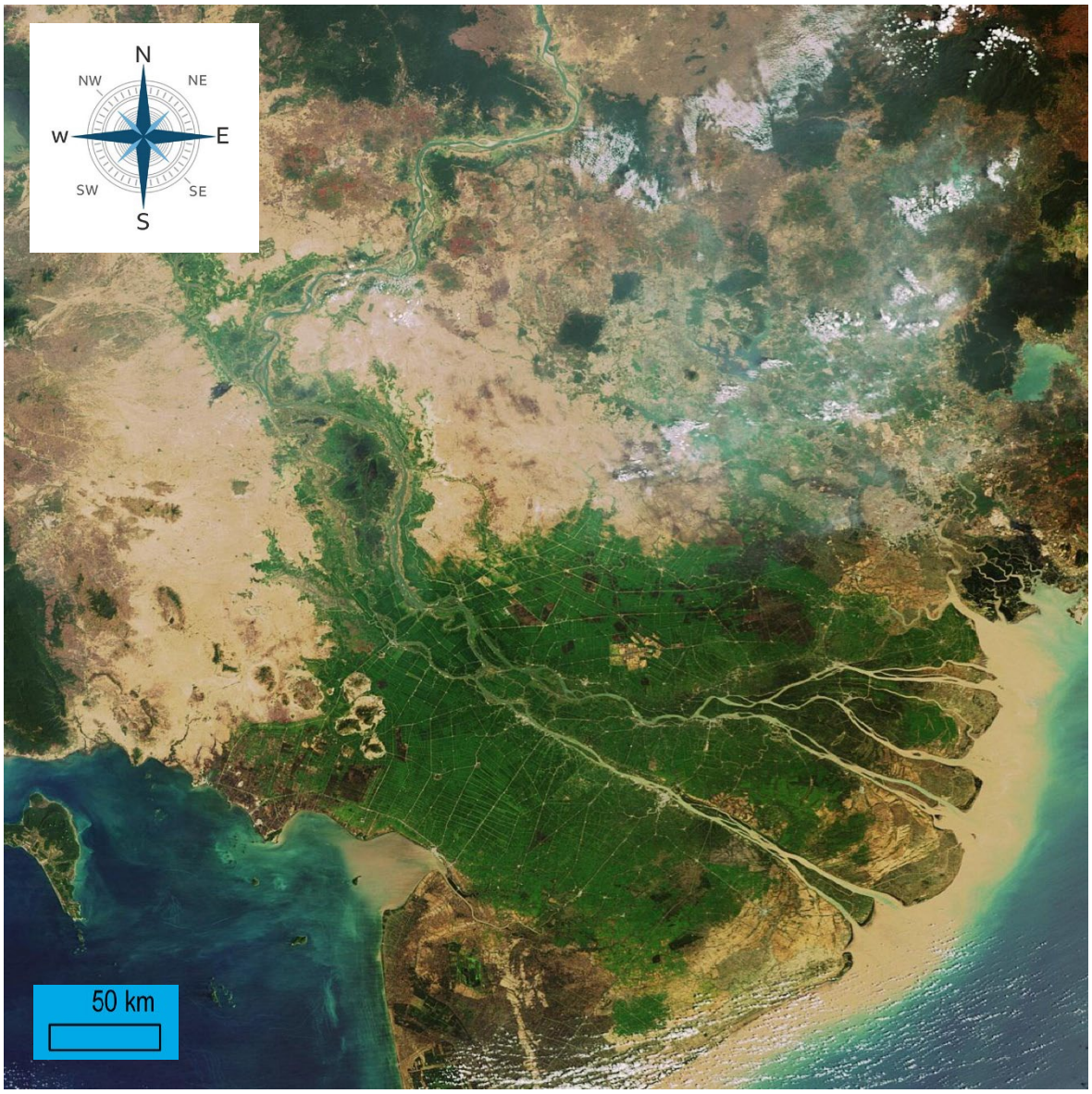
Global concern over China's control of critical mineral exports has intensified as carmakers and governments warn of supply disruptions. Beijing's April 4 curbs on seven rare earth elements (REEs) and related magnets - vital for vehicles, defense, and clean energy - require special export licenses, citing national security and non-proliferation. China produces 60% of the world's REEs and controls 90% of refining capacity (IEA, USGS 2025).

The restrictions are prompting urgent diplomatic and industry interventions. Japan, India, and European officials are seeking meetings to expedite licenses, while the U.S. - which sources over 61% of its rare earths from China - faces potential shortages. Carmakers including Tesla, GM, Toyota, and Volkswagen warn of production risks for components from sensors to power steering. The U.S. mine at Mountain Pass remains reliant on Chinese processing, and the new Texas facility will produce just 1,000 tonnes annually by 2027 versus China's 300,000 tonnes.

The issue has reached the highest diplomatic levels, with a planned call between Presidents Trump and Xi, amid U.S. accusations that China is violating the Geneva trade agreement by delaying export relief.

Resource 4 for Question 2

Satellite image of the Mekong Delta, Vietnam



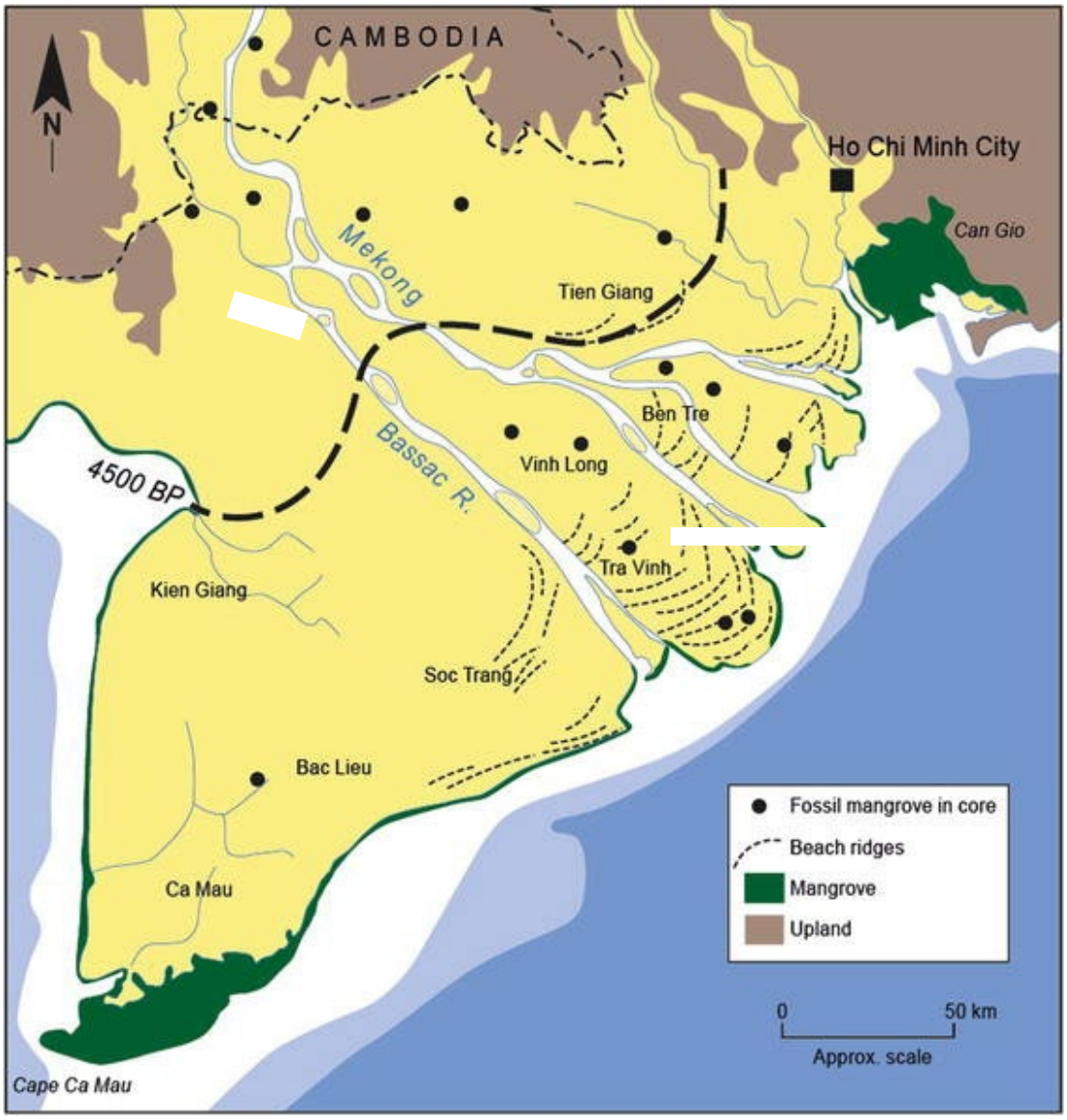
Resource 5 for Question 2

Location of the Mekong Delta, Vietnam



Resource 6 for Question 2

Distribution of mangrove forests on the Mekong Delta



The dotted line shows the coastline of the Mekong Delta 4500 years before present day.

Resource 7 for Question 2

Can Gio mangrove forest



The Can Gio mangrove forest contains a high diversity of animal species including the king cobra, saltwater crocodile and spot-billed pelican. It functions as the 'green lungs' of the neighboring Ho Chi Minh City by absorbing carbon dioxide. The forest also limits damage caused by storms and floods, reducing up to 50% of the impact from waves and sea level rise.

Sources

Resource 1	https://www.visualcapitalist.com/rare-earth-elements-where-in-the-world-are-they/
Resource 2	https://www.americasquarterly.org/article/rare-earth-elements-are-trending-will-they-drive-latin-americas-next-mining-boom/
Resource 3	Adapted from https://economictimes.indiatimes.com/industry/auto/china-rare-earth-minerals-export-xi-jinping-global-supply-chain-electric-vehicles-impact/articleshow/121614452.cms?from=mdr
Resource 4	https://www.esa.int/Applications/Observing_the_Earth/Earth_from_Space_Vietnam_s_nine-tailed_dragon
Resource 5	https://cdn.britannica.com/90/183790-050-7FC21C89/World-Data-Locator-Map-Vietnam.jpg
Resource 6	https://www.researchgate.net/figure/The-Mekong-River-Delta-showing-schematically-the-present-distribution-of-mangroves-and_fig1_300333171
Resource 7	https://i.pinimg.com/736x/9a/be/d7/9abed7056272085f7c3ad25308dbd58b.jpg https://www.unesco.org/en/mab/can-gio-mangrove